JOHN AARON MENDOZA BRANZUELA

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Computing Science student at Singapore Institute of Technology × University of Glasgow (graduating April 2027) who has built and deployed a production LLM assistant with RAG, locally hosted models and AI safety guardrails, with further experience in machine learning and cloud-native systems. Seeking to build secure, reliable and scalable AI solutions that people can trust.

EDUCATION

Bachelor of Science with Honours in Computing Science — Sep 2024 – Apr 2027
Singapore Institute of Technology × University of Glasgow

Diploma in Electrical Engineering with Eco-Design — Apr 2018 – Mar 2021
Nanyang Polytechnic · Director's List, SEM 1 AY2020/2021 (top-performing students)

WORK EXPERIENCE

LGA Telecom | Service Desk Operations & AI Integration Intern — Aug 2026 – Present

AI-Powered Service Desk Assistant (Desk Assist)

- **Reduced average ticket handling time** for Level 1 agents by building an LLM-powered assistant that analyses tickets and drafts responses, integrating Freshservice, n8n workflow orchestration, Ollama (DeepSeek-R1 14B, Qwen3 8B) and PostgreSQL conversation memory
- **Improved context-aware ticket resolution by 30%** (agent feedback surveys) through a Manifest V3 Chrome extension with knowledge-base retrieval, conversation analysis and AI-generated draft responses
- **Maintained 95% response quality** by implementing retrieval-augmented generation (RAG) over the company knowledge base, prompt engineering workflows and AI safety guardrails against sensitive-data leakage
- **Achieved zero data leakage incidents** by deploying on-premises Docker infrastructure with Nginx reverse proxy, TLS termination, port-based routing and FortiGate firewall IP allowlisting
- **Made the AI stack reproducible and observable** by orchestrating Ollama, n8n, OpenWebUI, PostgreSQL and Nginx with Docker Compose, version-controlling prompt templates and workflow configurations, and monitoring model availability and API response times
- **Cut MTTR by 15 minutes per incident** by integrating Freshservice with Microsoft Teams for automated escalation alerts, using webhook authentication and header-based credentials

FreeSDN Network Management Platform Integration

- **Enabled unified operations across 4+ equipment manufacturers** by leading full-stack integration and designing a vendor-agnostic architecture for provisioning, monitoring and troubleshooting
- **Migrated 1,000+ network configurations and monitoring baselines** via RESTful API integrations and migration scripts, and mapped 50+ existing workflows into a production cutover roadmap with acceptance criteria

Process Automation & Stakeholder Communication

- **Eliminated 4 hours of manual reporting per week** by automating PagerDuty incident log extraction and scheduled report delivery with Google Apps Script
- **Reduced manual agent actions by 40%** by designing Freshservice business rules and workflow automations for ticket closure, escalation routing and response tracking
- **Accelerated technical decisions for 10+ stakeholders** with SOPs, architecture diagrams, security assessments and ROI-focused proposals

Sembcorp Solar Singapore | Digitization Analyst Intern — Oct 2020 – Jun 2021

- **Prepared the Momens CRM platform for go-live** across SolarNova 3 and multiple divisions by designing and executing UAT, tracking 50+ defects to resolution
- **Delivered 20+ vendor-implemented enhancements** by running bi-weekly requirements sessions with cross-functional teams and consolidating inputs into formal change requests
- **Enabled company-wide CRM adoption** across three divisions with process guides and training materials

PROJECTS

SilverLink SG | React, Next.js, Node.js, gRPC, Kubernetes, PostgreSQL, Prisma, WebRTC **2025**

- **Built a cloud-native platform connecting seniors and youth** through skill-sharing and live events, with 4 gRPC microservices behind a REST-to-gRPC API gateway and WebRTC livestreaming; 6-person team, 7 Agile sprints
- **Scaled services ahead of demand** on an 8-node Kubernetes cluster using predictive autoscaling that forecasts load with an exponential moving average
- **Prevented double-booking under concurrent traffic** with version-checked optimistic concurrency and idempotency keys for exactly-once request handling, across 8 isolated per-service database schemas

SCS Awareness Platform | React 19, Supabase (PostgreSQL, Auth, Storage), Vercel **2025**

- **Built an interactive cancer-awareness experience for the Singapore Cancer Society**, a branching visual-novel engine deployed on tablets at community outreach booths in English, Chinese, Malay and Tamil
- **Owned the core engine and media systems**: panel renderer, multi-sprite character rendering, question bank service with editor validation across 4 question types, and admin authentication
- **Gave non-technical SCS staff a no-code admin panel** for stories, assets and analytics with CSV export, protected by Supabase Row-Level Security and a PDPA consent gate; 4-person team, 7 Agile sprints

UWB Indoor Localization | Python, scikit-learn, Pandas, NumPy, Matplotlib **2025**

- **Achieved 89.61% accuracy and 0.9612 ROC-AUC** with an SVM-RBF classifier on 41,996 samples from 7 indoor environments, after comparing candidate models
- **Improved model inputs** by applying PCA to 1,016 channel impulse response dimensions and engineering features to compensate for non-line-of-sight signal loss

TECHNICAL SKILLS

AI & Machine Learning: LLMs, Generative AI, RAG, Prompt Engineering, AI Safety Guardrails, Ollama, n8n, scikit-learn, Deep Learning, PyTorch (basic)

Programming & Data: Python, SQL, JavaScript/TypeScript, Pandas, NumPy, Matplotlib, PostgreSQL, ETL, Power BI

Cloud & DevOps: Docker, Docker Compose, Kubernetes, gRPC, Nginx, TLS/SSL, FortiGate, Git, Supabase, Vercel

Web: React, Next.js, Node.js, Express, Prisma ORM, WebRTC

Methods: Agile/Scrum, UAT/SIT, Requirements Gathering, Process Documentation, Stakeholder Engagement

CERTIFICATIONS

AI4I – Literacy in AI, AI Singapore (2026) · **Fundamentals of Deep Learning**, NVIDIA DLI (2026)

Kubernetes Fundamentals and **Docker Fundamentals**, Dell Technologies (2025)

HONOURS & LEADERSHIP

Internship Distinction, Sembcorp Solar Singapore (2021): led the CRM rollout; earned a contract extension

NYP JumpStart Grant Winner (SGD 10,000), GreenLoopFarms (2022): pitched to an industry investor panel

Student Facilitator, University of Glasgow Hackathon (2026): ran logistics for 70+ attendees